# Order of Operations

**Parentheses first. Then × and ÷. Then + and −.**

| 1. ( ) | 2. × ÷ | 3. + − |
|---|---|---|
| Parentheses first | Times and divide, left to right | Plus and minus, left to right |

**(8 + 4) × 3**

$(8 + 4) \times 3 = 12 \times 3 = 36$

The parentheses go first: 8 + 4 = 12.

**8 + 4 × 3**

$8 + 4 \times 3 = 8 + 12 = 20$

No parentheses: times goes first.

## Words to numbers

**“Add 5 and 7, then multiply by 2”** $(5 + 7) \times 2 = 24$

The part you do first goes in parentheses.

**Your turn!**

$(10 - 4) \times 5 =$ ☐

# Order of Operations: Practice

Parentheses first. Then × and ÷. Then + and −.

## Part 1 Parentheses first

Do the parentheses, then the rest.

1) $(3 + 4) \times 2 = \square \times 2 = \square$

2) $(15 - 6) \times 4 = \square \times 4 = \square$

3) $36 \div (2 + 7) = 36 \div \square = \square$

4) $(9 + 11) \div 5 = \square \div 5 = \square$

5) $6 \times (13 - 5) = 6 \times \square = \square$

6) $(24 - 3) \div 7 = \square \div 7 = \square$

7) $50 - (8 + 12) = 50 - \square = \square$

8) $(7 + 5) \times 8 = \square \times 8 = \square$

## Part 2 Parentheses

No help this time.

9) $(9 - 4) \times 6 = \square$

10) $40 \div (2 + 6) = \square$

11) $(15 + 9) \div 4 = \square$

## Order of Operations: Practice (continued)

### Part 2 Parentheses (continued)

12) $7 \times (8 - 3) =$ ☐

13) $(6 + 6) \times 5 =$ ☐

14) $(30 - 12) \div 3 =$ ☐

15) $9 \times (2 + 7) =$ ☐

16) $(45 - 5) \div 8 =$ ☐

17) $100 - (25 + 35) =$ ☐

18) $(14 + 6) \times 3 =$ ☐

19) $72 \div (12 - 4) =$ ☐

20) $(11 - 2) \times 9 =$ ☐

### Part 3 Times and divide first Do × or ÷, then the rest.

21) $5 + 3 \times 4 = 5 +$ ☐ $=$ ☐

22) $20 - 12 \div 3 = 20 -$ ☐ $=$ ☐

## Order of Operations: Practice (continued)

### Part 3 Times and divide first (continued)

23) $6 \times 7 + 8 = \square + 8 = \square$

24) $30 \div 5 - 2 = \square - 2 = \square$

25) $9 + 8 \times 2 = 9 + \square = \square$

26) $40 - 6 \times 5 = 40 - \square = \square$

27) $48 \div 6 + 15 = \square + 15 = \square$

28) $17 + 36 \div 4 = 17 + \square = \square$

### Part 4 Times and divide first No help this time.

29) $4 + 6 \times 3 = \square$

30) $50 - 8 \times 4 = \square$

31) $7 \times 6 - 12 = \square$

32) $45 \div 9 + 11 = \square$

33) $3 + 64 \div 8 = \square$

34) $9 \times 9 - 30 = \square$

## Order of Operations: Practice (continued)

### Part 4 Times and divide first (continued)

35) $100 - 56 \div 7 =$ ☐

36) $8 + 7 \times 7 =$ ☐

37) $63 \div 7 + 26 =$ ☐

38) $25 - 4 \times 5 =$ ☐

39) $12 \times 3 + 14 =$ ☐

40) $60 - 42 \div 6 =$ ☐

### Part 5 Same numbers, different answers

No help this time. Parentheses change the order.

41) $(2 + 3) \times 4 =$ ☐ and $2 + 3 \times 4 =$ ☐

42) $(10 - 2) \times 3 =$ ☐ and $10 - 2 \times 3 =$ ☐

43) $(18 + 6) \div 6 =$ ☐ and $18 + 6 \div 6 =$ ☐

44) $5 \times (4 + 2) =$ ☐ and $5 \times 4 + 2 =$ ☐

## Order of Operations: Practice (continued)

**Part 6** Words to numbers No help this time. Write the expression, then work it out.

45) **Add 6 and 9, then multiply by 3:** ☐

46) **Take 8 from 20, then divide by 4:** ☐

47) **Multiply 7 by 5, then add 12:** ☐

48) **Add 14 and 22, then divide by 6:** ☐

49) **Divide 81 by 9, then take away 4:** ☐

50) **Take 3 from 11, then multiply by 7:** ☐

**Part 7** How many times as big? No help this time. Don't work it out.

51) $4 \times (27 + 35)$ is ☐ times as big as $27 + 35$

52) $6 \times (189 + 52)$ is ☐ times as big as $189 + 52$

53) $3 \times (418 + 296)$ is ☐ times as big as $418 + 296$

54) $8 \times (75 + 64)$ is ☐ times as big as $75 + 64$

## Order of Operations: Practice (continued)

### Part 8 Parentheses first

55) $(3 + 4) \times 6 =$ ☐

56) $(12 + 3) \times 2 =$ ☐

57) $(11 + 4) \times 3 =$ ☐

58) $(9 + 8) \times 7 =$ ☐

59) $(11 + 8) \times 4 =$ ☐

60) $(7 + 9) \times 9 =$ ☐

61) $(8 + 11) \times 7 =$ ☐

62) $(5 + 10) \times 5 =$ ☐

63) $(5 + 14) \times 3 =$ ☐

64) $(15 + 4) \times 4 =$ ☐

65) $(4 + 11) \times 8 =$ ☐

66) $(10 + 11) \times 2 =$ ☐

## Order of Operations: Practice (continued)

### Part 8 Parentheses first (continued)

67) $(3 + 12) \times 6 =$ ☐

68) $(7 + 9) \times 2 =$ ☐

69) $(12 + 12) \times 9 =$ ☐

70) $(10 + 8) \times 7 =$ ☐

71) $(5 + 4) \times 7 =$ ☐

72) $(7 + 8) \times 8 =$ ☐

### Part 9 Times first

73) $40 + 3 \times 7 =$ ☐

74) $22 + 6 \times 7 =$ ☐

75) $3 + 6 \times 8 =$ ☐

76) $36 + 3 \times 2 =$ ☐

77) $22 + 5 \times 6 =$ ☐

78) $3 + 8 \times 8 =$ ☐

## Order of Operations: Practice (continued)

### Part 9 Times first (continued)

79) $6 + 5 \times 4 =$ ☐

80) $40 + 6 \times 7 =$ ☐

81) $22 + 4 \times 3 =$ ☐

82) $17 + 8 \times 7 =$ ☐

83) $20 + 9 \times 9 =$ ☐

84) $34 + 6 \times 9 =$ ☐

85) $14 + 7 \times 5 =$ ☐

86) $33 + 5 \times 3 =$ ☐

87) $39 + 3 \times 8 =$ ☐

88) $32 + 5 \times 7 =$ ☐

89) $8 + 3 \times 3 =$ ☐

90) $9 + 4 \times 8 =$ ☐

## Order of Operations: Practice (continued)

### Part 10 Divide first

91) $21 - 27 \div 3 =$ ☐

92) $40 - 36 \div 4 =$ ☐

93) $59 - 21 \div 7 =$ ☐

94) $40 - 40 \div 5 =$ ☐

95) $54 - 6 \div 3 =$ ☐

96) $26 - 72 \div 8 =$ ☐

97) $35 - 56 \div 8 =$ ☐

98) $20 - 40 \div 5 =$ ☐

99) $59 - 8 \div 4 =$ ☐

100) $29 - 32 \div 4 =$ ☐

101) $23 - 15 \div 3 =$ ☐

102) $35 - 45 \div 9 =$ ☐

## Order of Operations: Practice (continued)

### Part 11 Remember? Problems from earlier packets.

103) $\frac{1}{8} \div 3 = \frac{\square}{\square}$

104) $5 \div \frac{1}{3} = \square$

105) $6 \div 8 = \frac{\square}{\square}$

106) $7 \times 0.9 = \square$

107) 5 cm × 2 cm × 6 cm = ☐ cubic cm

108) 3.6 to the nearest whole number = ☐

### Part 12 Word problems Write one expression, then work it out.

109) Mia buys 3 packs of 8 pens and 5 more pens. How many pens does she have?

Number sentence _____

Answer ☐ pens

Draw or show your work

110) Leo has 40 stickers. He gives away 12 and shares the rest among 4 friends. How many each?

Number sentence _____

Answer ☐ stickers

Draw or show your work

111) A ticket costs $6. 4 adults and 5 kids go. How much do they pay in all?

Number sentence _____

Answer ☐

Draw or show your work

## Order of Operations: Practice (continued)

### Part 12 Word problems (continued)

112) Zoe has 50 beads. She uses 6 beads on each of 7 bracelets. How many beads are left?

Number sentence _____

Answer ☐ beads

Draw or show your work

113) Challenge: Ben reads 15 pages a day for 4 days, then 25 pages. Sam reads 80. Who reads more?

Number sentence _____

Answer ☐

Draw or show your work

# Check-In: Order of Operations

Show what you know. No help this time. Score: _____ / 8

## Part 1 Parentheses

1) $(20 - 11) \times 6 =$ ☐

2) $(64 - 16) \div 6 =$ ☐

3) $(33 + 7) \div 5 =$ ☐

## Part 2 Times and divide first

4) $70 - 8 \times 8 =$ ☐

5) $8 \times 4 - 9 =$ ☐

6) $2 + 9 \times 6 =$ ☐

## Part 3 Words to numbers

7) **Take 6 from 30, then divide by 8:** ☐

8) **Multiply 8 by 6, then take away 10:** ☐